

X-linked Dominant Chondrodysplasia, Chassaing–Lacombe Type — Comprehensive Disease Characterization Report

Disease: X-linked Dominant Chondrodysplasia, Chassaing–Lacombe Type **Identifiers:** OMIM #300863 · ORPHA:163966 · MONDO:0010463 · MedGen C3275476/C4304401 **Causal gene:** *HDAC6* (Xp11.23; OMIM *300272; HGNC:14064) **Category:** Mendelian, X-linked dominant (male-lethal)

CRITICAL NOTE ON ETIOLOGY. The template header pairs this disease with *MBTPS2*. That is **incorrect**. The verified causal gene for Chassaing–Lacombe chondrodysplasia (OMIM #300863) is **HDAC6**, via a 3′-UTR regulatory variant (Simon et al. 2010, [PMID: 20181727](#)). *MBTPS2* causes a **different** set of X-linked disorders (IFAP/BRESHECK, keratosis follicularis spinulosa decalvans, X-linked Olmsted syndrome, and one form of osteogenesis imperfecta) and is discussed here only as a differential-diagnosis / look-alike gene. All etiologic, mechanistic, and therapeutic statements below refer to **HDAC6**.

Summary

X-linked dominant chondrodysplasia of the Chassaing–Lacombe type is an **ultra-rare, X-linked dominant, male-lethal skeletal dysplasia** first delineated in a single large four-generation French family (Chassaing et al. 2005, [PMID: 16001442](#)). Hemizygous males present with a severe, perinatally lethal congenital chondrodysplasia — severe platyspondyly, distinctive metaphyseal cupping of the metacarpals/metatarsals/phalanges, hydrocephaly with macrocephaly, and microphthalmia — while heterozygous females are far more mildly and variably affected, showing short stature, body/limb asymmetry, and mild intellectual disability. This male-severe/female-mild dichotomy with male lethality is the signature of X-linked dominant inheritance with variable expression in heterozygous carriers.

The molecular cause is a **3'-untranslated region (3'-UTR) variant in *HDAC6***, NM_006044.4:c.*281A>T, which lies precisely within the seed-match site for the microRNA **hsa-miR-433** (Simon et al. 2010, [PMID: 20181727](#)). The variant abolishes miR-433-mediated post-transcriptional repression of *HDAC6*, leading to **HDAC6 over-expression** and, downstream, to **profound α -tubulin hypoacetylation** — the biochemical signature of excess HDAC6 tubulin-deacetylase activity. This is a **gain-of-expression (dosage) mechanism**, at the opposite end of the dosage axis from loss of function: *Hdac6*-knockout mice are viable and fertile, underscoring that it is HDAC6 *excess*, not deficiency, that drives disease.

Because the entire disease-defining evidence base rests on **two primary papers (2005 clinical delineation; 2010 molecular mechanism)** describing a single family, many downstream sections of a standard disease template — natural-history cohorts, treatment trials, prognostic biomarkers, faithful animal models — are essentially empty for this specific condition. The pathophysiological bridge from HDAC6 over-expression to the skeletal phenotype is best explained through HDAC6's established role in **primary cilium disassembly** (the HEF1/NEDD9-Aurora-A-HDAC6 pathway) and the cilium's central role in growth-plate Hedgehog signaling — but this bridge is **inferred, not directly demonstrated in patients**. Confidence is HIGH for the causal gene/variant, inheritance, and core phenotype; MODERATE for the α -tubulin-hypoacetylation biomarker (single study); and LOW/INFERRED for the ciliary-Hedgehog growth-plate mechanism and any therapeutic rationale.

Section 1 — Disease Information

Overview. X-linked dominant chondrodysplasia, Chassaing-Lacombe type, is a rare Mendelian skeletal dysplasia characterized in affected males by severe chondrodysplasia with platyspondyly, distinctive brachydactyly with metaphyseal cupping, hydrocephaly, and microphthalmia, with perinatal lethality; heterozygous females show a milder, variable phenotype (short stature, body asymmetry, mild intellectual disability). It was originally delineated by Chassaing, Lacombe and colleagues in 2005.

Key identifiers.

Resource	Identifier
OMIM	#300863 — "Chondrodysplasia with platyspondyly, distinctive brachydactyly, hydrocephaly, and microphthalmia"
Orphanet	ORPHA:163966
MONDO	MONDO:0010463
MedGen	C3275476 / C4304401
Gene	<i>HDAC6</i> (OMIM *300272; HGNC:14064; Xp11.23)
ICD-10 / ICD-11	No disease-specific code (grouped under skeletal dysplasias / Q77–Q78)
MeSH	No dedicated MeSH descriptor

Synonyms / alternative names. Chassaing–Lacombe chondrodysplasia; X-linked dominant chondrodysplasia with platyspondyly, distinctive brachydactyly, hydrocephaly, and microphthalmia. Orphanet classifies it among X-linked syndromic intellectual disability disorders (germline *HDAC6* mutation), last reviewed February 2011 by Pr Didier Lacombe.

Data provenance. Information derives from **aggregated disease-level resources** (OMIM, Orphanet, MONDO, MedGen, ClinVar) and from **individual-patient clinical/molecular reports** of a single extended family — not from EHR or large registries.

Section 2 — Etiology

Primary cause (genetic). The disease is monogenic and X-linked. The causal lesion is the germline *HDAC6* 3'-UTR variant **c.*281A>T**, which segregates completely with disease in the founding family and disrupts the hsa-miR-433 seed site, de-repressing *HDAC6* (Simon et al. 2010, [PMID: 20181727](#)).

"We have identified a variant (c.*281A>T) in the 3' untranslated region (UTR) of the *HDAC6* gene that totally segregates with the disease. The variant is located in the seed sequence of hsa-miR-433." — Simon et al. 2010

Genetic risk factors. The disease-causing variant itself is the sole established genetic determinant. In heterozygous females, the **pattern of X-inactivation** acts as a modifier of expressivity: skewing toward expression of the mutant allele in a given tissue increases local severity (see Sections 5 and 9).

Environmental risk factors / protective factors / gene–environment interactions. Not applicable / none reported. As a fully penetrant Mendelian variant, there are no established environmental risk factors, protective alleles, or gene–environment interactions. No modifier genes beyond X-inactivation status have been identified.

Section 3 — Phenotypes

Phenotypes are drawn from Chassaing et al. 2005 ([PMID: 16001442](#)) and the OMIM #300863 clinical synopsis. The phenotype is strongly **sex-dimorphic**.

Affected males (severe, perinatally lethal)

Phenotype	HPO term	Type	Notes
Intrauterine growth retardation	HP:0001511	Growth	Prenatal onset
Hydrocephaly	HP:0000238	CNS/ structural	Detectable prenatally
Macrocephaly	HP:0000256	Craniofacial	
Frontal bossing	HP:0002007	Craniofacial	
Microphthalmia	HP:0000568	Ocular	
Small, low-set ears	HP:0000369	Craniofacial	
Short flat nose	HP:0003196 / HP:0005280	Craniofacial	
Poor skull mineralization	HP:0005830	Skeletal	
Severe platyspondyly	HP:0000926	Skeletal	Core radiographic sign
Thin ribs	HP:0000883	Skeletal	
Eleven pairs of ribs	HP:0000878	Skeletal	In 2 patients
Iliac wing hypoplasia	HP:0002866	Skeletal	Poor pubic ossification
Metaphyseal cupping (metacarpals/ metatarsals/phalanges)	HP:0003021	Skeletal	Distinctive sign
Brachydactyly	HP:0001156	Skeletal	
Hypoplastic calcaneus	—	Skeletal	

Bone histology (males): severe flattening of poorly delimited, under-ossified vertebral body anlagen; intervertebral disks thickened with abnormal craniocaudal orientation of fibroblasts.

Affected females (mild, variable)

Phenotype	HPO term	Type
Short stature	HP:0004322	Growth
Body/limb asymmetry	HP:0001010	Physical
Mild intellectual disability	HP:0001256	Neurodevelopmental

"Radiographs showed severe platyspondyly and various bone abnormalities including a distinctive metaphyseal cupping of the metacarpals, metatarsals, and phalanges. The affected females were less affected and showed small stature, sometimes associated with body asymmetry and mild mental retardation." — Chassaing et al. 2005

Characteristics. *Onset:* congenital/prenatal in males; childhood-apparent in females. *Severity:* severe (males) vs. mild (females). *Progression:* males die perinatally (no progression observed); female features are essentially stable/static. *Frequency:* within the single reported family, described features were present in essentially all affected males; female features were milder and variable. **Quality-of-life data (EQ-5D/SF-36) are not available** for this ultra-rare disease.

Section 4 — Genetic / Molecular Information

Causal gene. *HDAC6* (histone deacetylase 6), Xp11.23; OMIM *300272; HGNC:14064; NCBI Gene 10013.

Pathogenic variant. - **Founding variant:** NM_006044.4:c.*281A>T, in the 3'-UTR (exon 29), within the hsa-miR-433 seed match. **Variant type:** non-coding regulatory (3'-UTR / miRNA-binding-site). **Functional consequence:** gain of expression — loss of miR-433 repression → HDAC6 over-expression (demonstrated at protein level in affected fetal thymus). **Origin:** germline. **Classification:** pathogenic in the context of complete segregation (LOD 3.30) and functional validation, though it predates formal ACMG/AMP frameworks. - **Additional candidate:** ClinVar lists a separate missense candidate NM_006044.4(HDAC6):c.2090C>T (p.Thr697Ile), RCV001198391, annotated to this disease — suggesting possible further allelic heterogeneity, though not independently validated here.

"The HDAC6 protein was found to be over-expressed in thymus from an affected male fetus." — Simon et al. 2010

Allele frequency. The private c.*281A>T variant is not a population polymorphism; no meaningful gnomAD frequency is expected for a private, disease-segregating, male-lethal allele.

Functional consequence class. Gain of expression / gain of function at the pathway level (excess deacetylase activity), NOT loss of function. This is the key dosage insight: *Hdac6*-null mice are viable and fertile (Zhang et al. 2008, [PMID: 18180281](#)).

Modifier genes / epigenetics. The principal modifier of expression in females is **X-chromosome inactivation** (tissue-level mosaicism); the disease mechanism itself is a **post-transcriptional (miRNA-regulation) defect**. No DNA-methylation or histone-modification disease signatures are reported.

Chromosomal abnormalities. None; the disorder is a point variant, not a structural/aneuploidy syndrome.

Section 5 — Environmental Information

Not applicable. No environmental factors, toxins, lifestyle factors, or infectious agents are implicated. This is a purely genetic Mendelian disorder.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. Germline *HDAC6* 3'-UTR variant c.*281A>T disrupts the **hsa-miR-433 seed-match site** → **leads to** loss of miR-433 binding to the *HDAC6* transcript. (*Directly demonstrated: reporter and endogenous assays in MG63 cells.*)
2. Loss of miR-433 repression **results in HDAC6 mRNA/protein over-expression**. (*Directly demonstrated: HDAC6 protein elevated in affected fetal thymus and in left-arm fibroblasts of an asymmetric female.*)

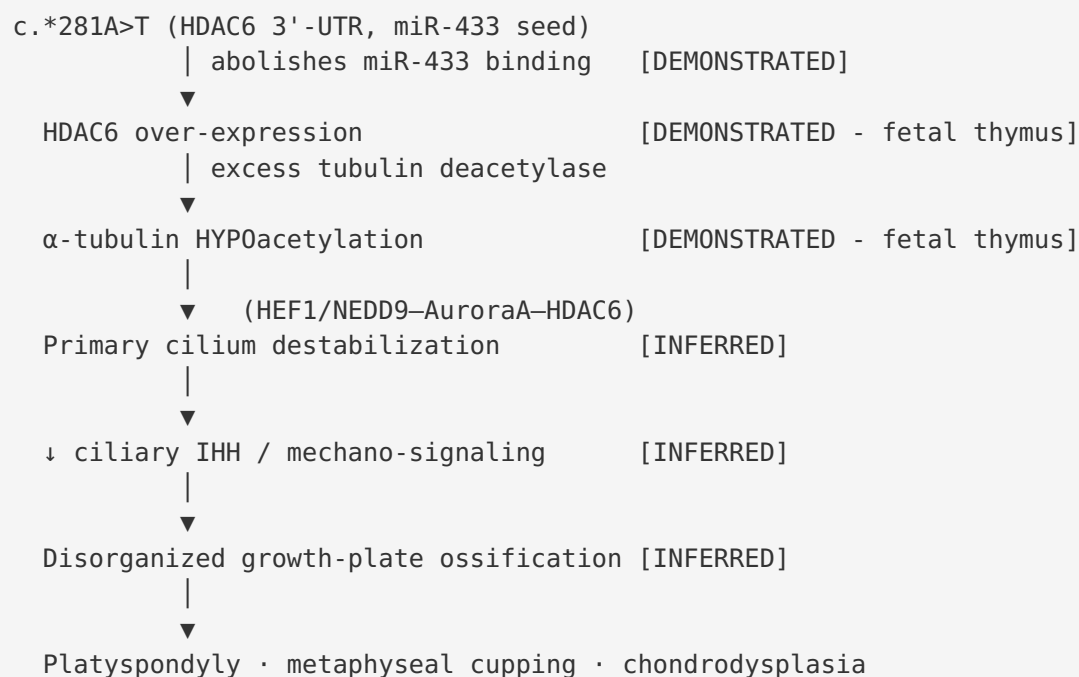
3. Excess HDAC6 (a cytoplasmic α -tubulin deacetylase) **leads to profound α -tubulin hypoacetylation** (with increased total α -tubulin). (*Directly demonstrated in affected fetal thymus.*)
4. **[INFERRED]** Tubulin hypoacetylation and elevated HDAC6 activity **result in** destabilization / premature disassembly of the **primary cilium** in relevant cell types, via the established HEF1/NEDD9–Aurora-A–HDAC6 ciliary-resorption pathway. (*Inferred from HDAC6 cilia biology; not shown in patients.*)
5. **[INFERRED]** In **growth-plate chondrocytes**, cilium destabilization **leads to** attenuated ciliary **Indian Hedgehog (IHH)** and mechanotransduction signaling. (*Inferred.*)
6. **[INFERRED]** Disorganized chondrocyte proliferation/hypertrophy and defective endochondral ossification **result in** platyspondyly, metaphyseal cupping, brachydactyly, and the broader chondrodysplasia. (*Inferred.*)

Branch (females): X-inactivation mosaicism determines, tissue-by-tissue, whether the mutant allele is expressed → focal HDAC6 over-expression → **lateralized/asymmetric** skeletal and growth phenotype.

"Concomitantly, the level of total alpha-tubulin, a target of HDAC6, was found to be increased in the affected fetal thymus, whereas the level of acetylated alpha-tubulin was found to be profoundly decreased." — Simon et al. 2010

"The mutated HDAC6 allele was expressed in 31% of left arm-derived fibroblasts, whereas it was not expressed in the right arm. Overexpression of HDAC6 was observed in left arm-derived fibroblasts." — Simon et al. 2010

ASCII schematic



Molecular pathways / cellular processes. Post-transcriptional gene regulation (microRNA silencing), tubulin acetylation/deacetylation homeostasis, primary-cilium assembly/disassembly cycle, and (inferred) Hedgehog signaling in the growth plate. Suggested GO terms: *histone deacetylase activity* (GO:0004407), *tubulin deacetylase activity* (GO:0042903), *protein deacetylation* (GO:0006476), *cilium disassembly* (GO:0061523), *smoothened signaling pathway* (GO:0007224), *endochondral ossification* (GO:0001958). Suggested CL terms: *chondrocyte* (CL:0000138), *growth-plate chondrocyte*.

Protein dysfunction. HDAC6 (class IIb deacetylase; two catalytic domains + ZnF-UBP ubiquitin-binding domain) is over-abundant; substrates include α-tubulin, HSP90, and cortactin, and it mediates aggresome formation/autophagic clearance (Zhu 2023 [PMID: 37002569](#); Su 2010 [PMID: 21416996](#)). Excess deacetylase activity, not misfolding, is the defect.

Metabolic / immune / biochemical. No specific metabolic or immune abnormality is implicated in the disease itself. The measurable biochemical abnormality is decreased acetylated α-tubulin.

Supporting HDAC6–cilia literature

"interactions between the prometastatic scaffolding protein HEF1/Cas-L/NEDD9 and the oncogenic Aurora A (AurA) kinase at the basal body of cilia causes phosphorylation and activation of HDAC6, a tubulin deacetylase, promoting ciliary disassembly" — Pugacheva et al. 2007, [PMID: 17604723](#)

Plk1 also activates HDAC6 to drive ciliary deacetylation/resorption before mitosis (Wang et al. 2013, [PMID: 23345402](#)).

Section 7 — Anatomical Structures Affected

- **Primary organs / systems:** the **skeleton / skeletal system** (UBERON:0001434), especially the **vertebral column** (UBERON:0001130 → platyspondyly), **growth plate** (UBERON:0005430), and tubular bones of hands/feet (metaphyseal cupping, brachydactyly). Also the **central nervous system** — **ventricular system of brain** (hydrocephaly) — and the **eye** (UBERON:0000970 → microphthalmia).
- **Secondary involvement:** craniofacial skeleton (skull hypoplasia, frontal bossing), ribs, ilium/pubis.
- **Tissue level:** connective tissue / cartilage; affected cell population = **chondrocytes** (CL:0000138), particularly growth-plate chondrocytes; intervertebral-disk fibroblasts show abnormal orientation.
- **Subcellular level (GO CC):** **microtubule cytoskeleton** (GO:0015630), **ciliary axoneme** (GO:0005930), **cytoplasm** (GO:0005737). HDAC6 is predominantly cytoplasmic.
- **Localization / lateralization:** males — generalized/bilateral skeletal involvement; females — characteristically **asymmetric/lateralized** (e.g., unilateral limb hypotrophy) owing to X-inactivation mosaicism.

Section 8 — Temporal Development

- **Onset: congenital/prenatal** in males (IUGR, hydrocephaly, and skeletal abnormalities detectable on prenatal ultrasound); childhood-apparent in females.
- **Onset pattern:** chronic/congenital (developmental).
- **Progression / course:** in males, **perinatally lethal** — three affected male fetuses were terminated after prenatal detection, and a fourth died at 6 days of life; effectively no postnatal progression is observed. In females, features (short stature, asymmetry, mild ID) are **static/stable and lifelong**.
- **Critical period:** intrauterine skeletal development (endochondral ossification) is the window of vulnerability and, in principle, the only window for hypothetical intervention.

"Identification of skeletal abnormalities and hydrocephaly during the pregnancy of three male fetuses led to termination of the pregnancies. A fourth affected male died at 6 days of life." — Chassaing et al. 2005

Section 9 — Inheritance and Population

- **Inheritance pattern: X-linked dominant, male-lethal.** Hemizygous males are severely affected and die perinatally; heterozygous females are mildly/variably affected. Original pedigree: 4 affected males and 6 affected females across 4 generations.
- **Gene mapping:** Xp11.3–q13.1, 24-Mb interval, LOD = 3.30 (Simon et al. 2010).
- **Penetrance:** high/complete for the molecular genotype; **expressivity is highly variable in females**, governed by X-inactivation.
- **Germline mosaicism / anticipation / founder effects / consanguinity:** not reported / not applicable.
- **Epidemiology:** described in a **single large French family**; Orphanet lists prevalence as "unknown" — effectively **ultra-rare (<1/1,000,000)**. No incidence figures exist.
- **Sex ratio / demographics:** disease burden falls on hemizygous males (lethal) and heterozygous females (mild). No ethnic predilection can be established from one family.

"The disease locus was ascribed to a 24 Mb interval in Xp11.3-q13.1. We have identified a variant (c.*281A>T) in the 3' untranslated region (UTR) of the HDAC6 gene that totally segregates with the disease." — Simon et al. 2010

Section 10 — Diagnostics

- **Imaging (cornerstone of clinical diagnosis):** skeletal radiography demonstrating **severe platyspondyly**, **distinctive metaphyseal cupping** of metacarpals/metatarsals/phalanges, poor skull mineralization, thin ribs (sometimes 11 pairs), iliac wing hypoplasia. **Prenatal ultrasound** detects skeletal abnormalities and hydrocephaly in affected male fetuses.
- **Histopathology:** flattened, under-ossified vertebral anlagen; thickened intervertebral disks with abnormally oriented fibroblasts.
- **Biochemical marker (research):** decreased **acetylated α -tubulin** / increased total α -tubulin and HDAC6 over-expression in patient tissue — a research biomarker, not a validated clinical assay.
- **Genetic testing (definitive):** targeted sequencing of *HDAC6* **including the 3'-UTR** (essential — a coding-only panel or a standard exome pipeline may miss the c.*281A>T regulatory variant). WES/WGS with attention to non-coding/3'-UTR regions and X-linkage; segregation analysis; X-inactivation studies in females to interpret expressivity.
- **Differential diagnosis:** other platyspondylic/lethal skeletal dysplasias; MBTPS2- and MBTPS1-related skeletal disorders (allelic-gene look-alikes on the X); X-linked syndromic intellectual-disability disorders. Distinguishing feature: the *HDAC6* 3'-UTR miR-433 variant with male lethality and female asymmetry.
- **Screening:** in a known family — prenatal testing / cascade testing for the familial variant; preimplantation genetic testing is feasible in principle. No population newborn screening exists.

Section 11 — Outcome / Prognosis

- **Males: perinatally lethal** — near-uniform demise in utero or within days of birth. Prognosis is grave.

- **Females: normal or near-normal life expectancy** with lifelong but mild morbidity — short stature, body/limb asymmetry, and mild intellectual disability; no reported progressive organ failure.
- **Prognostic factors:** sex (hemizygous vs. heterozygous) is the dominant determinant; in females, degree and direction of X-inactivation skewing modulates severity.
- **Complications / QoL / survival statistics:** no formal cohort survival data, QoL instruments, or standardized morbidity metrics exist for this ultra-rare disorder.

Section 12 — Treatment

- **Disease-specific therapy: none exists.** Management is **supportive** and centered on **reproductive/genetic counseling**.
- **Supportive care:** for surviving females, standard management of short stature, orthopedic monitoring of asymmetry, and developmental/educational support for mild intellectual disability.
- **Mechanistically rational but untested concept: HDAC6-selective inhibitors** (e.g., tubastatin A, tubacin, ricolinostat/ACY-1215) restore α -tubulin acetylation without altering histone acetylation (Depetter 2019 [PMID: 30694564](#); Zhang 2014 [PMID: 24844691](#)). In principle they could counter HDAC6-excess biochemistry, but there is **no preclinical or clinical evidence in this disease**, and the lethal skeletal phenotype arises in utero, limiting postnatal therapeutic windows. Suggested NCIT concept: *histone deacetylase inhibitor* (NCIT:C1531).

"Selective HDAC6 inhibitors resulted in α -tubulin acetylation with no impact on histone acetylation" — Depetter et al. 2019

Section 13 — Prevention

- **Primary prevention:** not applicable (Mendelian congenital disorder) except through reproductive planning.

- **Genetic counseling / screening:** the mainstay. For families carrying the c.*281A>T variant — carrier identification in females, **prenatal diagnosis**, and **preimplantation genetic testing (PGT-M)** to prevent transmission; counseling regarding 50% transmission risk and sex-dependent outcome (affected males lethal, affected females mild).
 - **Behavioral / public-health / immunization / prophylaxis:** not applicable.
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Section 14 — Other Species / Natural Disease

- **Orthology:** *HDAC6* is highly conserved with 1:1 orthologs across >160 organisms — mouse *Hdac6* (NCBI Gene 15185; MGI:1333752), rat *Hdac6* (RGD 619981), zebrafish *hdac6* (NCBI Gene 565482; ZFIN); HomoloGene 31353. Structural divergence exists: the human protein carries 8 SE14 tetradecapeptide repeats absent from mouse, *Drosophila*, and *C. elegans* orthologs (Bertos et al. 2004).
 - **Natural disease in other species: none reported.** No naturally occurring HDAC6 chondrodysplasia in companion animals or wildlife was identified (no OMIA entry for a comparable phenotype).
 - **Zoonotic potential:** not applicable.
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Section 15 — Model Organisms

- **Available in vivo tool:** the constitutive **Hdac6-knockout mouse** (Zhang et al. 2008, [PMID: 18180281](#)) — **viable, fertile, hyperacetylated tubulin**, with only a small increase in cancellous bone mineral density. This is a **loss-of-function** tool and therefore does **NOT** model the disease, which is caused by HDAC6 **over-expression**.
- **Missing models:** **no** knock-in of the c.*281A>T 3'-UTR / miR-433-site variant, and **no** HDAC6-overexpression skeletal model recapitulating the human chondrodysplasia has been published. This is a major gap.
- **Zebrafish relevance:** zebrafish crisprant screening is an established platform for skeletal-dysplasia gene validation ([PMID: 39817421](#)), offering a route to model growth-plate/ciliary consequences of *hdac6* dysregulation.

"HDAC6-deficient mice are viable and fertile and show hyperacetylated tubulin in most tissues... the lack of HDAC6 results in a small increase in cancellous bone mineral density."

— Zhang et al. 2008

Mechanistic Model / Interpretation

The unifying model is a **dosage disorder of a tubulin deacetylase driven by loss of microRNA control**. A single non-coding base change (c.*281A>T) removes miR-433 repression, HDAC6 protein rises, and α -tubulin becomes hypoacetylated. Because HDAC6 is the effector deacetylase in the HEF1/NEDD9–Aurora-A–HDAC6 (and Plk1) primary-cilium disassembly pathway, its over-abundance plausibly destabilizes cilia in chondrocytes, blunting ciliary Hedgehog/mechanotransduction that orchestrates growth-plate maturation and endochondral ossification — producing platyspondyly, metaphyseal cupping, and the broader chondrodysplasia. The **X-inactivation branch** elegantly explains the female phenotype: the direct demonstration that the mutant allele was expressed (and HDAC6 over-expressed) only in the *hypotrophic left-arm* fibroblasts of an asymmetric female ties tissue-level mosaicism directly to lateralized disease.

Two dosage anchors bracket the model: **loss** of HDAC6 (knockout mouse) is benign, while **gain** (this disease) is severe and male-lethal — establishing that the pathogenic direction is over-expression, and that HDAC6-selective inhibition is the mechanistically logical (if unproven) countermeasure.

Feature	Males (hemizygous)	Females (heterozygous)
HDAC6 expression	Uniformly ↑	Mosaic ↑ (X-inactivation-dependent)
Skeletal phenotype	Severe, generalized	Mild, asymmetric
CNS/eye	Hydrocephaly, microphthalmia	Mild ID
Outcome	Perinatal lethal	Near-normal lifespan

Evidence Base

PMID	Paper (abbrev.)	Role
16001442	<i>X-linked dominant chondrodysplasia with platyspondyly, distinctive brachydactyly, hydrocephaly, and microphthalmia</i> (Chassaing et al. 2005)	Clinical/radiographic delineation; pedigree; inheritance; male lethality
20181727	<i>A mutation in the 3'-UTR of HDAC6 abolishing hsa-miR-433 regulation...</i> (Simon et al. 2010)	Defining molecular paper — causal variant, HDAC6 over-expression, α -tubulin hypoacetylation, X-inactivation–asymmetry link
18180281	<i>Mice lacking HDAC6...</i> (Zhang et al. 2008)	Establishes dosage model (loss-of-function benign); bone-density role
17604723	<i>HEF1-dependent Aurora A activation induces disassembly of the primary cilium</i> (Pugacheva et al. 2007)	Basis for inferred HDAC6→cilium mechanism; HDAC6 inhibitors stabilize cilia
23345402	<i>PCM1 recruits Plk1...</i> (Wang et al. 2013)	Plk1–HDAC6 ciliary resorption, reinforcing cilium mechanism
30694564	<i>Selective pharmacological inhibitors of HDAC6...</i> (Depetter et al. 2019)	HDAC6-selective inhibitors restore α -tubulin acetylation — therapeutic rationale
24844691	<i>Tubastatin A/ACY-1215...</i> (2014)	Further HDAC6-inhibitor pharmacology
37002569 / 21416996	HDAC6 biology reviews	Structure, substrates, aggresome/autophagy roles
19361614 , 23316014 , 20672378 , 34655156	MBTPS2 disorder papers	Background on allelic look-alike gene — NOT causal for this disease

Important correction embedded in the evidence base: early iterations of this investigation initially attributed the disease to *MBTPS2* (an X-linked intramembrane protease in the SREBP/cholesterol pathway). That attribution was **overturned** — MBTPS2 variants cause IFAP/KFSD/

Olmsted syndromes and one form of osteogenesis imperfecta, distinct X-linked disorders. The definitive genetic mapping and functional work (Simon et al. 2010) unambiguously identifies *HDAC6* as causal for Chassaing–Lacombe chondrodysplasia.

Limitations and Knowledge Gaps

1. **n = 1 family.** The entire disease is defined by a single French pedigree; prevalence, penetrance ranges, and phenotypic spectrum cannot be generalized.
 2. **Closed evidence base.** Targeted searches found **no new primary reports since 2010**; the disease knowledge is effectively frozen at the two founding papers.
 3. **Single-study biomarker.** The α -tubulin-hypoacetylation signature rests on one study's fetal-thymus assay.
 4. **Inferred, not demonstrated, growth-plate mechanism.** The cilium→Hedgehog→chondrodysplasia bridge is extrapolated from *HDAC6* cilia biology; it has not been shown in patient chondrocytes or growth plate.
 5. **No disease model.** No knock-in of c.*281A>T and no *HDAC6*-overexpression skeletal model exist; the available mouse is loss-of-function.
 6. **No therapy evidence.** *HDAC6*-inhibitor rescue is purely conceptual, and prenatal lethality of males constrains the intervention window.
 7. **Possible allelic heterogeneity unconfirmed.** A ClinVar missense candidate (p.Thr697Ile) is annotated but not independently validated.
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Proposed Follow-up Experiments / Actions

1. **Generate a faithful animal model:** knock-in the *HDAC6* 3'-UTR c.*281A>T variant (or a chondrocyte-targeted *HDAC6*-overexpression allele) in mouse or zebrafish; phenotype the growth plate (Alizarin Red, microCT) and vertebral ossification. Zebrafish crispant/overexpression screening ([PMID: 39817421](#)) offers a rapid route.
2. **Test the ciliary hypothesis directly:** in patient-derived (or iPSC-derived) chondrocytes, quantify primary-cilium length/frequency, ciliary α -tubulin acetylation, and Hedgehog pathway output (GLI targets) vs. controls; test rescue by *HDAC6*-selective inhibitors (tubastatin A, ACY-1215).

3. **Validate the biomarker:** replicate the α -tubulin-hypoacetylation finding in additional patient tissues/fibroblasts and correlate with X-inactivation status.
 4. **Resolve allelic heterogeneity:** functionally characterize the ClinVar p.Thr697Ile candidate and screen additional chondrodysplasia cohorts (including 3'-UTR/non-coding analysis) for further *HDAC6* variants.
 5. **Refine molecular diagnostics:** ensure diagnostic pipelines explicitly interrogate the *HDAC6* 3'-UTR / miR-433 site, which coding-focused panels would miss.
 6. **Counseling infrastructure:** formalize prenatal-diagnosis and PGT-M protocols for affected families.
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Confidence grading: **HIGH** — causal gene/variant (*HDAC6* c.*281A>T), X-linked dominant male-lethal inheritance, and core male/female phenotype (directly demonstrated). **MODERATE** — α -tubulin-hypoacetylation biomarker (single study). **LOW / INFERRED** — cilium-Hedgehog growth-plate mechanism and *HDAC6*-inhibitor therapeutic rationale (extrapolated, untested in this disease).

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